Instrumentación II: *Adquisición de datos*

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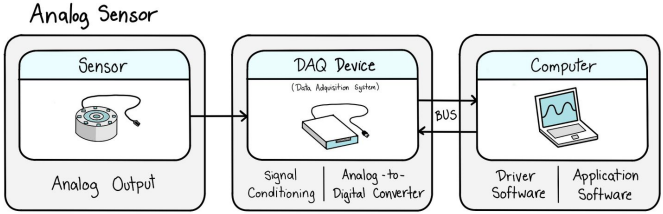
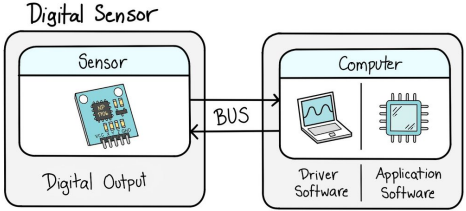
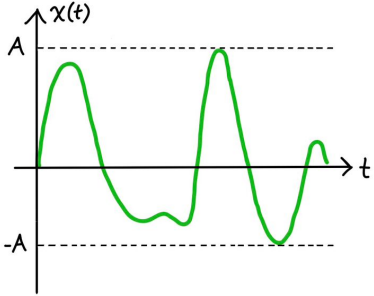
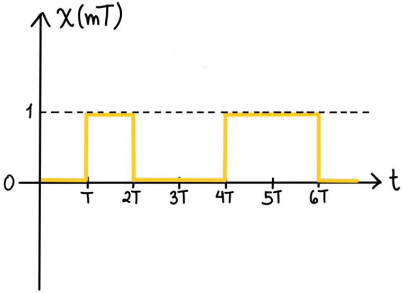
Sistema de adquisición de datos



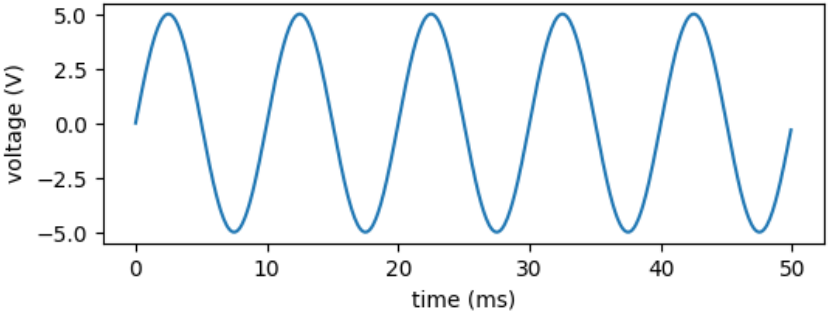


Señales

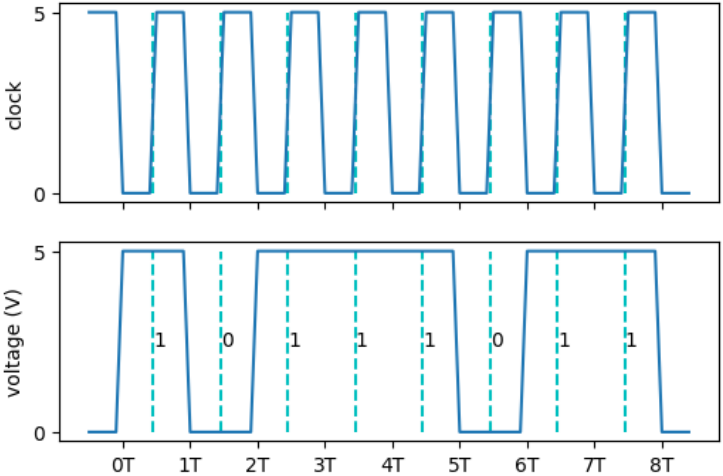




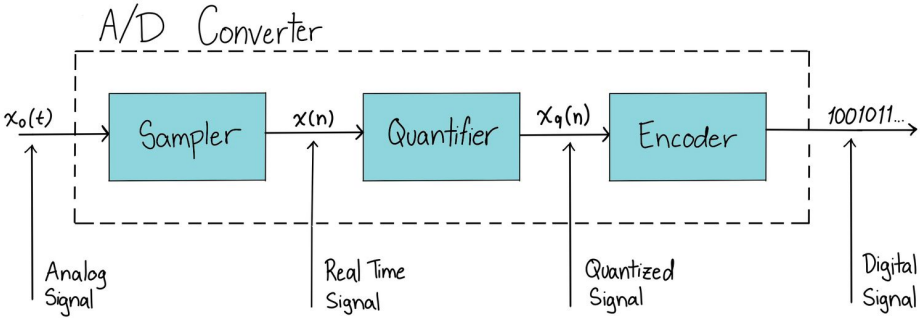
Analógica



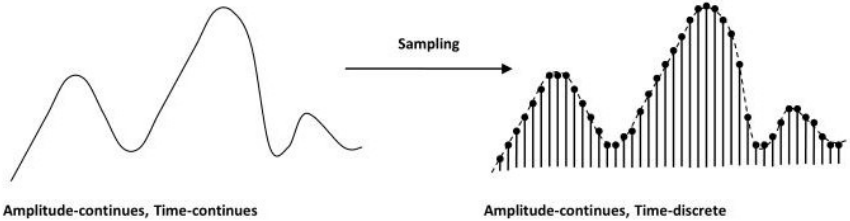
Digital

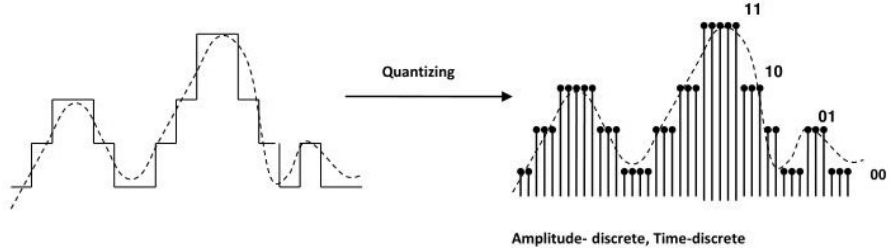


Etapas de la conversión A/D



Muestreo, cuantización y codificación



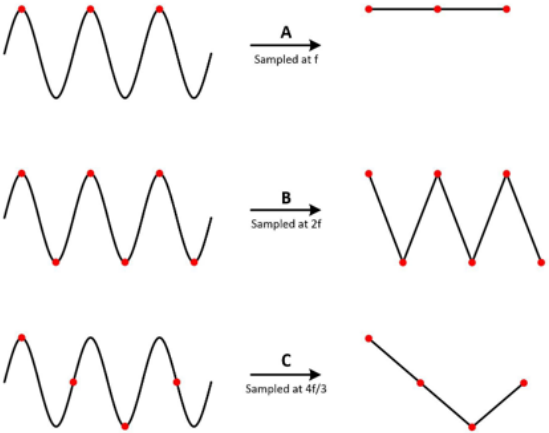


Muestreo

Frecuencia de Nyquist

$$f_N \geq 2 \cdot f_{max}$$

una frecuencia de muestreo muy baja puede causar una reconstrucción inexacta de la forma de onda



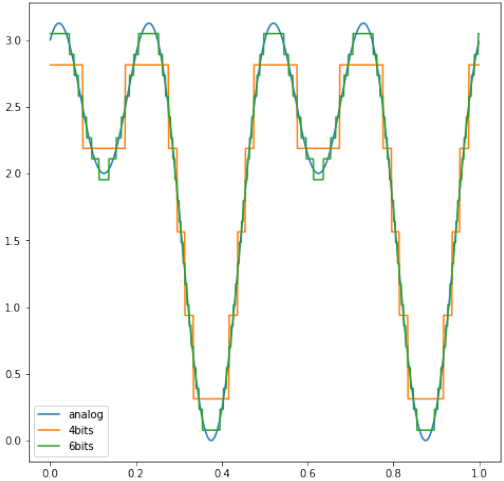
Cuantización

Resolución

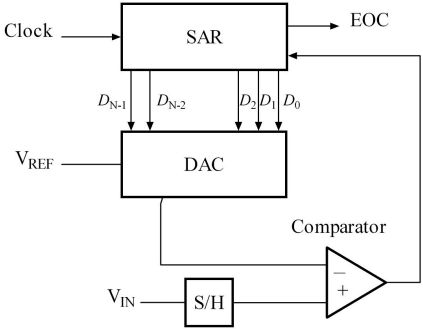
$$\Delta = \frac{\text{Rango}}{2^b - 1}$$

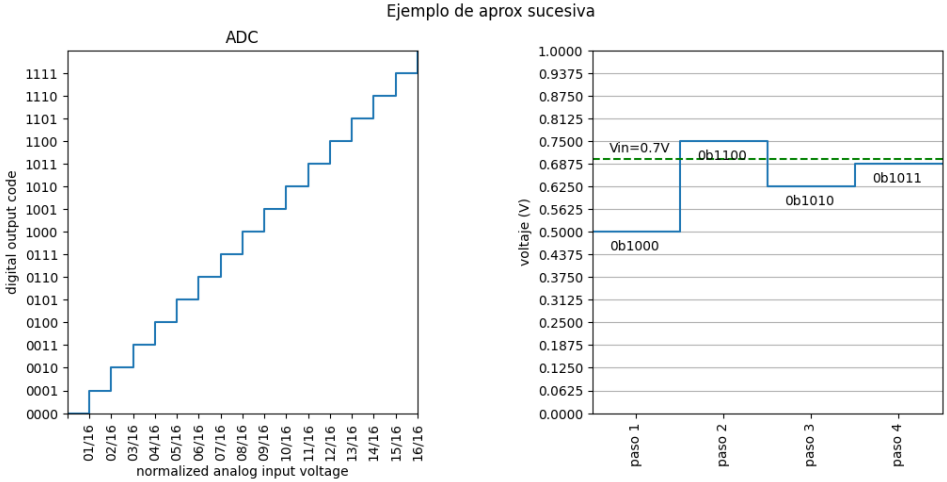
Error de cuantización

$$|e_q| \leq \frac{\Delta}{2}$$



Ejemplo: aproximación sucesiva



Sinha, Aritra and Sunit Kumar Sen. “Design of an improved successive approximation type ADC using multi bit per cycle algorithm for conversion rate improvement.” Proceedings of The 2014 International Conference on Control, Instrumentation, Energy and Communication (CIEC) (2014): 219-223.

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